PROJECT-BASED LEARNING

PHASE-II EVALUATION

RakshaNetra – NHAI Worker Safety and Emergency Response System

Team ID: DSCPP-III-2026-T072

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team Information

Team ID	DSCPP-III-2026-T072
Semester	3rd
Section	DS1, AIML1, AIML2, AIML3
Domain	AIML and AIDS

Mentor

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Phase-II: System Development and Feature Implementation

Problem Statement

NHAI construction sites involve many workers, staff, vehicles, and locations, making safety records difficult to manage manually.

During emergencies, finding worker details and arranging suitable rescue support can take valuable time.

RakshaNetra provides a centralized system for worker management, identity verification, safety incident recording, and emergency response.

Motivation

- Manual and scattered records.
- Delayed emergency response.
- No single safety system.
- Need for quick access to worker information.

Stakeholders

- NHAI site staff and workers.
- Rescue teams and site managers.

Project Objectives

1. Develop a simple and organized system to manage worker safety and emergency situations at NHAI construction sites.
2. Manage worker records, verification, safety incidents, and emergencies.
3. Integrate Data Structures in C and Object-Oriented Programming in C++.
4. Implement practical modules for worker management and emergency response.
5. Test the system with different worker, safety, and emergency situations.

Key Objective

Improve NHAI worker safety through quick identification and emergency response.

Proposed Approach

1. Collect worker, staff, site, and emergency details.
2. Verify workers and assess emergencies.
3. Record safety incidents and maintain digital history.
4. Prioritize emergencies according to severity.
5. Store and display required safety information.

Core Features

- Worker and Staff Record Management
- Worker Identity Verification
- Safety Incident Recording
- Emergency Severity and Priority Management
- Rescue Team Selection and Assignment

The Phase-II implementation focuses on converting the proposed architecture into functional safety and emergency-management modules.

01 Worker and Staff Record Management

Core Functionality

- Centralized digital storage of all worker and staff details.
- Stores: Worker ID, Name, Role, Contact, Blood Group, Site Location, and Join Date.
- Supports complete record operations: Add, Update, Search, Delete, and Display.
- Eliminates manual registers and reduces data loss.
- Provides quick retrieval of worker information during emergencies.

Implementation

C Programming
Arrays + Linked Lists

Record Operations
Add → Search → Update → Delete → Display

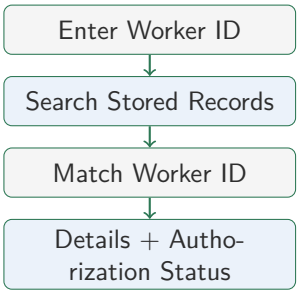
Emergency Benefit
Fast access to worker information

02 Worker Identity Verification

Core Functionality

- Instant verification of workers and staff using a unique Worker ID.
- Prevents unauthorized entry on NHAI construction sites.
- Matches the entered Worker ID with stored records in real-time.
- Displays worker details and authorization status.
- Improves site security and accountability.

Verification Flow



Searching Techniques in C: Linear Search / Binary Search

03 Safety Incident Recording

Core Functionality

- Allows staff to record any safety incident or near-miss immediately.
- Captures: Incident Type, Location, Time, Description, and Involved Workers.
- Maintains a complete history of safety events.
- Helps identify repeated issues and improve safety measures.
- Creates a permanent digital safety log for the site.

Incident Record

Incident Type
Accident / Near-Miss

Location
Construction Site

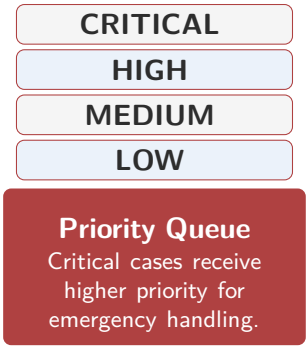
Time
Incident Timestamp

04 Emergency Severity and Priority Management

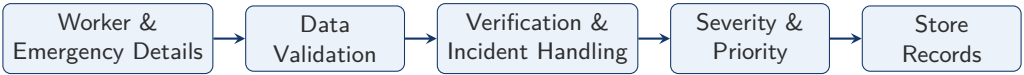
Core Functionality

- Staff can register emergencies and assign severity levels.
- Severity levels: Low → Medium → High → Critical.
- System automatically prioritizes critical cases using a Priority Queue.
- Higher severity emergencies are handled first.
- Reduces response time during accidents.
- Displays the priority list of pending emergencies for quick action.

Emergency Priority Queue



System Architecture / Workflow



Worker Management

- Worker records
- Staff records
- Identity verification

Safety Management

- Incident recording
- Near-miss logging
- Safety history

Emergency Management

- Severity levels
- Priority Queue
- Emergency response

Worker data → Processing → Storage → Safety / Emergency Output

DSA & OOP Mapping to System Modules

Concept	Implementation	System Usage
Arrays	C	Worker record storage
Linked Lists	C	Dynamic worker and incident records
Linear / Binary Search	C	Worker identity verification
Queues / Priority Queue	C	Emergency prioritization
Structures	C	Worker and incident data organization
Classes and Objects	C++	Rescue team management
Inheritance	C++	Reusable management structures
Polymorphism	C++	Flexible rescue team operations

Technology Stack

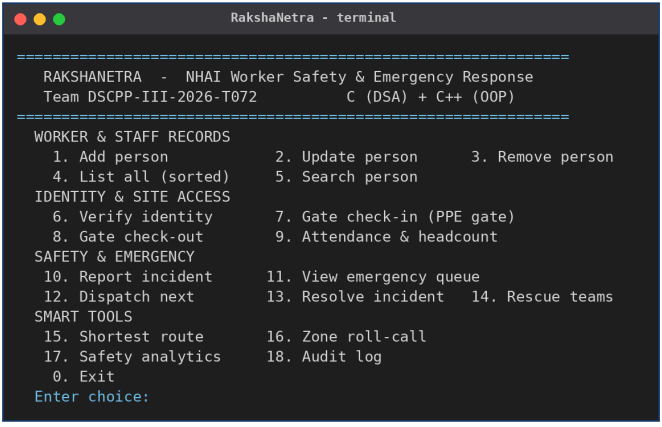
- **Programming:** C and C++
- **Database:** MySQL
- **Tools / Framework:** STL
- **IDE:** Visual Studio Code
- **Version Control:** Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C and C++ support the Data Structures and OOP concepts used in RakshaNetra, while MySQL supports organized data storage and retrieval.

Implementation: Application Interface



Main menu of RakshaNetra (console interface).

What the interface offers

- 18 menu options in four groups.
- Worker and staff records.
- Identity and site access.
- Safety and emergency handling.
- Smart tools: route, roll-call, analytics, audit log.

Implementation: Worker and Staff Records

RakshaNetra - terminal

```
Enter choice: 4
--- All Workers & Staff ---
  ID    Name              Role        Task                Zone            Auth  Status
  ------------------------------------------------------------------------------------
  104   Aarti Negi        Worker      Material Handling   Material Yard   Yes   ON-SITE
  108   Anil Thapa        Staff       Safety Officer      Admin Office    Yes   ON-SITE
  106   Deepak Bisht      Worker      Crane Signalman     Bridge Pier A   Yes   ON-SITE
  111   Imran Ali         Worker      Truck Driver        Main Gate       NO    off-site
  110   Kavita Joshi      Contractor  Survey Contractor   Material Yard   Yes   ON-SITE
  103   Mohan Singh       Worker      Tunnel Drilling     Tunnel Section  Yes   ON-SITE
  112   Neha Rawat        Staff       Camp Nurse          Medical Camp    Yes   ON-SITE
  107   Priya Sharma      Staff       Site Engineer       Admin Office    Yes   ON-SITE
  ...
  Total records: 12
```

List of all workers, staff and contractors sorted by name (linked list + insertion sort in C). Role names come from C++ polymorphism.

Implementation: Identity Verification and PPE Gate

RakshaNetra - terminal

```
Enter choice: 6
--- Identity Verification ---
  Enter ID shown on card: 107
  Enter name on card: Priya Sharma
  RESULT: VERIFIED - genuine NHAI Staff (Blue badge), task: Site Engineer
Enter choice: 7
--- Gate Check-In (ID + PPE + Zone access) ---
  Scan / enter ID: 105
  ACCESS DENIED: missing PPE -> Boots
  (Worker must wear: Helmet Vest Boots)
Enter choice: 7
--- Gate Check-In (ID + PPE + Zone access) ---
  Scan / enter ID: 109
  ACCESS DENIED: Contractors cannot work alone in restricted zone [Tunnel Section].
Enter choice: 7
--- Gate Check-In (ID + PPE + Zone access) ---
  Scan / enter ID: 111
  ACCESS DENIED: ID card revoked.
```

Verification and gate check-in on live data.

Checks performed

- ID and name matched with records.
- Revoked ID cards are blocked.
- Entry denied if PPE is missing.
- Contractors blocked from restricted zones.

Implementation: Emergency Priority Queue

RakshaNetra - terminal

```
Enter choice: 11
--- Emergency Priority Queue (highest priority first) ---
  Rank #    Type      Severity  Location         Time     Description
  ------------------------------------------------------------------------------------------
  1    7    Fire      CRITICAL  Tunnel Section   16:02:30 Fire in tunnel
  2    4    Collapse  Severe    Excavation Zone  16:02:28 Trench wall partly collapsed, worker trapped
  3    5    Medical   Serious   Bridge Pier A    16:02:28 Worker dizzy at height, needs medical check
```

Pending emergencies, highest priority first. A newly reported critical fire jumps to rank 1 (max-heap).

RakshaNetra - terminal

```
Enter choice: 14
--- Rescue Teams ---
  ID  Team             Type           Base           Status
  ------------------------------------------------------------------
  1   Medical Alpha    Medical        Medical Camp   AVAILABLE
  2   Ambulance Bravo  Medical        Main Gate      AVAILABLE
  3   Fire Unit 1      Fire & Safety  Fire Post      BUSY
  4   Heavy Rescue 1   Heavy Rescue   Material Yard  BUSY
```

Implementation: Rescue Dispatch and Route

RakshaNetra - terminal

```
Enter choice: 12
--- Dispatch Next Emergency ---
  Highest priority: incident #7 - Fire / CRITICAL at Tunnel Section
  "Fire in tunnel"
  TEAM ASSIGNED: Fire Unit 1 (Fire & Safety), suitability 10/10
  Route: Fire Post -> Admin Office -> Material Yard -> Excavation Zone -> Tunnel Section
  Distance: 1000 m   Estimated arrival: ~5 min (at 250 m/min)
  Persons currently ON-SITE in [Tunnel Section]:
   - [103] Mohan Singh (Worker) - Tunnel Drilling
  Total: 1 person(s). Confirm every one is accounted for.
```

The top emergency is dispatched. The best team is chosen by suitability and distance, the shortest route (Dijkstra) is shown, and the people in the danger zone are listed.

Implementation: Safety Analytics and Audit Log

```
RakshaNetra - terminal

Enter choice: 17
--- Safety Analytics Dashboard ---
  Total incidents: 7    Average severity: 3.00 / 5
  Status -> Logged: 2  Pending: 2  Dispatched: 1  Resolved: 2
  Incidents by type:
   Medical    3 ###
   Fire       2 ##
   Vehicle    1 #
   Collapse   1 #
  Incidents by severity:
   1 (Minor   )  1 #
   2 (Low     )  1 #
   3 (Serious )  3 ###
   4 (Severe  )  1 #
   5 (CRITICAL)  1 #
  Zone risk ranking (most incidents first):
   1. Material Yard    2 incident(s)
   2. Tunnel Section   2 incident(s)
   3. Main Gate        1 incident(s)
   4. Bridge Pier A    1 incident(s)
  Recommendation: increase safety inspections at [Material Yard].
```

Safety analytics dashboard.

```
RakshaNetra - terminal

Enter choice: 18
--- Audit Log (last 12 events, oldest first) ---
   1. [16:02:33] System started. Sample data loaded.
   2. [16:02:34] Incident #7 Fire sev 5
   3. [16:02:34] DISPATCH #7 -> Fire Unit 1
```

Audit log (circular queue).

Progress Achieved

- NHAI worker safety issues identified.
- Worker and emergency requirements identified.
- C/C++ system architecture and workflow prepared.
- Worker record management module developed.
- Worker identity verification module developed.
- Safety incident recording module prepared.
- Emergency severity and priority management designed.

Individual Contribution

Member	Role	Contribution
Soumya Rawat	Lead	Planning and coordination
Sarthak Singh Chauhan	Developer	C/C++ module development
Pranjal Pandey	DB/Testing	Database and testing
Tazeen Anjum	Developer/Docs	Development and documentation

Roadmap

Phase-I
Proposal & System Design

Phase-II
Feature Development & Testing

Phase-III
Final Implementation & Evaluation

Expected Outcomes

- Working worker safety management system.
- Faster access to worker information.
- Improved worker identity verification.
- Permanent digital safety incident history.
- Priority-based emergency handling.
- Future expansion with real-time tracking and alerts.

References

1. NHAI Official Website
2. Road Safety Research Papers
3. C / C++ Documentation
4. MySQL Documentation

Project Focus

- Worker and staff management
- Identity verification
- Safety incident recording
- Emergency priority management
- Data Structures and OOP integration

Thank You

Questions & Suggestions

RakshaNetra – NHAI Worker Safety and Emergency Response System